Apollo 13 pogo: when propulsion and structure formed a feedback loop

Learning objective. Explain why the Apollo 13 S-II event was a coupled-system instability - not merely a high vibration environment - and translate that distinction into analysis, test and design-review decisions.

During ascent on April 11, 1970, the Saturn V's second stage developed 14-16 Hz longitudinal oscillations. Three episodes peaked near 180, 250 and 330 seconds after liftoff. The last episode became severe enough that liquid-oxygen (LOX) pump-discharge pressure variations actuated the center J-2 engine's thrust-OK pressure switches. The engine shut down at 330.65 seconds, 132.36 seconds earlier than planned [2].

LOCAL CENTER ENGINE	SPACECRAFT	MISSION RESPONSE
33.7 g measured/estimated at the thrust pad near cutoff; the 2008 NASA retrospective rounds this to 34 g and notes the accelerometer went out of range [2,3].	Less than 0.05 g at 16 Hz during S-II boost [1].	Four outboard engines burned 34.53 s longer; the S-IVB burn also lengthened, restoring near-nominal orbital insertion [1,2].

The contrast is the lesson. A vehicle-level sensor can remain benign while a localized structural-propulsion loop is racing toward loss of margin. The correct safety question is not only What acceleration reaches the payload? It is also Is any coupled loop capable of generating energy faster than the system dissipates it?

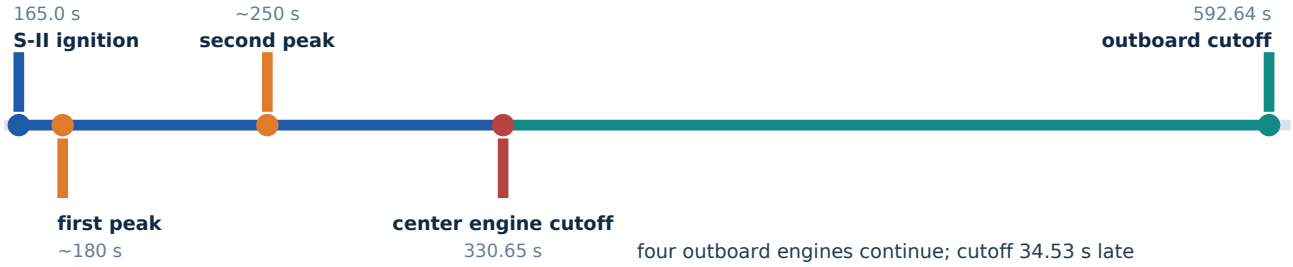
Evidence boundary

All event values above are reported by NASA sources. The growth-rate calculation later in this lesson is explicitly illustrative; it is not a reconstruction of Apollo 13 telemetry. NASA's 2008 statement that only one additional growth cycle might have been tolerable is a post-flight estimate, not a directly measured limit [3].

01 / THE EVENT

A local instability, an automatic shutdown, a recoverable trajectory

FIGURE 2 Original event timeline compiled from the AS-508 flight evaluation report [2].



The primary flight report describes three distinct structural/propulsion-coupled episodes. Earlier oscillations remained bounded; the third did not. Near cutoff, the center-engine thrust-pad response reached 33.7 g in a 12-18 Hz bandpass, while the center-engine thrust-chamber-pressure oscillation reached about 236 psi amplitude. A LOX-sump accelerometer recorded 5.7 g [2]. These numbers are local measurements with different transfer paths and cannot be compared as though they were colocated response channels.

The report also states that the oscillations decayed to normal after center-engine cutoff and that analysis found no structural failure. The guidance system then exploited engine-out capability: the four outboard engines continued to 592.64 seconds, 34.53 seconds later than predicted, and the next stage burned longer. The vehicle achieved a near-nominal parking orbit [1,2].

What failed - and what did not

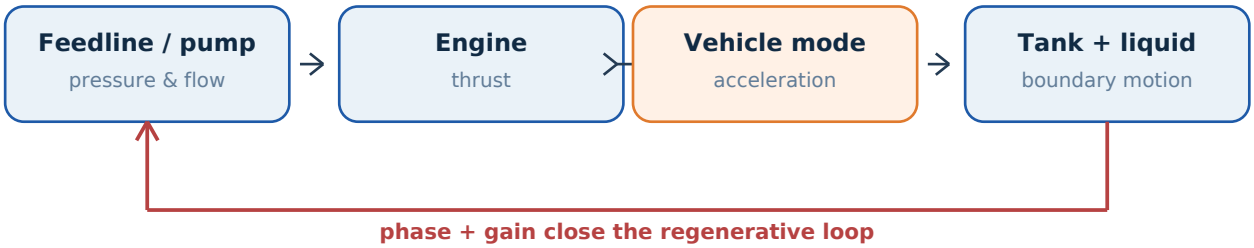
Function	Observed result	Reliability interpretation
Coupled stability	Failed locally; divergent S-II center-engine loop.	A stability deficiency existed even though the upper vehicle environment was modest.
Engine protection	Thrust-OK switches initiated cutoff.	Protection converted an escalating structural hazard into a known engine-out case.
Mission-level robustness	Remaining stages compensated.	Graceful degradation preserved ascent performance; it did not erase the initiating design hazard.

Safety significance. A protection trip is successful only if the trip logic, structural survival time, engine-out trajectory, residual propellant and downstream guidance authority remain compatible. Apollo 13 demonstrated that layered robustness can save a mission segment, but it also came uncomfortably close to the layer's limit [3].

02 / THE PHYSICS

Pogo is a feedback problem wearing a vibration signature

FIGURE 1 Original functional diagram; arrows denote small perturbations about the operating point.



A small longitudinal motion of the stage changes the inertial boundary conditions of the propellant and the pump. Feedline pressure and flow fluctuate; pump response converts inlet perturbations into discharge-pressure and flow perturbations; chamber pressure and thrust fluctuate; thrust then excites the same structural motion. Tank-bottom and local crossbeam motion close the loop. If the loop returns force with the phase that adds energy to the structural mode and the loop gain exceeds loss, effective damping becomes negative.

A compact frequency-domain description is

$$L(j\omega,t) = H_{structure} \cdot H_{tank/feed} \cdot H_{pump/engine} \cdot H_{thrust}$$
$$\text{closed-loop poles: } 1 - L(s,t) = 0$$

The explicit time argument matters. Propellant mass, ullage pressure, inlet pressure, cavitation compliance, structural frequencies and gains evolve through ascent. A loop can be stable at ignition, move toward neutral stability as modes approach, then restabilize after a crossing. Apollo 13's three episodes were therefore not three independent random shocks; they were manifestations of a moving coupled system [2,4].

Why apparently small changes matter

NASA SP-8055 records large flight-to-flight dispersions for nominally similar vehicles and warns that small parameter deviations can greatly change severity. Critical quantities include structural modal frequency, shape and damping; feedline acoustic modes; pump cavitation compliance and dynamic gain; accumulator compliance and inertance; engine gain; and the phase of each transfer path [4].

The center engine sat on a crossbeam and coupled strongly to the LOX tank-bottom/feed system. A remote command-module accelerometer had little observability of that local loop. Instrumentation placement must therefore follow modal participation and loop topology, not only crew or payload environment locations.

Established finding vs. hypothesis. The reports establish the 14-16 Hz coupled oscillation, local response, pressure-switch cutoff and post-cutoff decay. Exact nonlinear saturation mechanisms and the counterfactual time to failure are model-based interpretations; they should not be presented as measured telemetry.

03 / WORKED ILLUSTRATION

Half a percent of damping decides decay or divergence

For one lightly damped mode with natural frequency f_n, the slowly varying response envelope is

$$A(t) = A_0 \exp(-\zeta \omega_n t), \text{ where } \omega_n = 2\pi f_n$$

Use f_n = 16 Hz and compare ζ = +0.005 with ζ = -0.005. These values are invented to show sensitivity; they are not Apollo 13 damping estimates. For negative damping, the doubling time is

$$t_2 = \ln(2) / (|\zeta| 2\pi f_n) = 0.693 / (0.005 \times 2\pi \times 16) = 1.38 \text{ s} \approx 22 \text{ cycles}$$

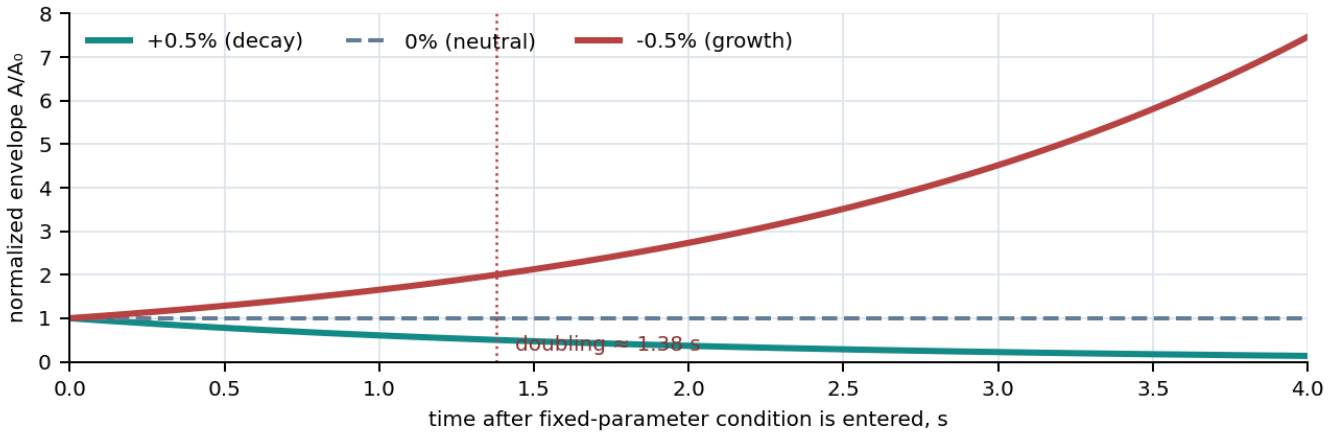


FIGURE 3 Original illustrative calculation. The plot holds frequency and damping fixed; a flight vehicle does neither. It demonstrates why a small sign error in net damping can matter within seconds.

The same 0.5% magnitude produces mirror-image behavior: one case dissipates energy and the other accumulates it. After 4 s, the idealized negative-damping envelope has grown by $\exp(0.005 \times 2\pi \times 16 \times 4) \approx 7.5$. Conversely, the positive-damping envelope falls to about 0.13. A loads-only process that assumes a finite stationary amplitude can completely miss this bifurcation.

Assumptions and limitations

This single-mode envelope assumes linear, time-invariant coefficients, light damping and free growth. Real pogo is multi-input/multi-output and time varying. Pump cavitation, feedline impedance, structural joints, actuator limits, shutdown logic and other nonlinearities can alter growth or create a limit cycle. NASA SP-8055 explicitly states that predicting nonlinear limit-cycle amplitude was beyond the supported state of the art; its design approach emphasized linear stability margin, uncertainty and test correlation instead [4].

04 / DESIGN RESPONSE

The program moved from accepting loads to eliminating instability

Before Apollo 13, operational mitigation shut the S-II center engine down 60-75 seconds early to avoid the worst predicted pogo region. A helium-bleed toroidal suppressor for the center-engine LOX line was available, but installation, inspection and verification concerns on the waiting AS-508 vehicle contributed to the decision to fly without it. The analysis predicted nonlinear amplitude limitation. Flight disproved that comfort: the third episode diverged [3].

The AS-508 anomaly report recommended adding an accumulator in the center-engine LOX feedline to lower the line natural frequency and decouple it from the crossbeam mode. It was installed on AS-509 (Apollo 14) and subsequent Apollo vehicles; NASA reports no further Apollo pogo beyond a brief end-of-burn buzz [2,3].

A practical VS&A; review matrix

Question	Evidence to demand	Why it protects reliability
Can a coupled pole cross into instability?	Closed-loop eigenvalues or Nyquist/root-locus results across flight time, with traceable subsystem models.	Separates self-excitation from a finite forcing environment.
What uncertainty controls margin?	Distributions/bounds for damping, modal gain/phase, pump dynamic gain, cavitation compliance and accumulator state.	Near-neutral loops amplify small parameter errors and tail combinations.
Will test observe the loop?	Synchronized local accelerations, pump-inlet/discharge pressures, chamber pressure and operating-state channels.	Remote response alone can miss a high-participation local mode.
What happens after detection?	Sensor range, trip voting/latency, structural survival time, engine-out trajectory and downstream control authority.	Makes protection and graceful degradation analytically closed.
Is the suppressor healthy?	Charge/fill verification, failure cases, modal separation and performance interaction.	A suppressor is part of the coupled plant, not an add-on vibration absorber.

Scope note. NASA SP-8055 is a 1970 design-criteria monograph, not a universal modern launch-vehicle requirement. Its enduring technical principle is to demonstrate coupled-system stability using analysis plus test, carry uncertainty, and modify the physical system when margin is inadequate [4]. Program-specific standards and hazard controls still govern today's implementation.

05 / RETAIN AND APPLY

Three takeaways

1. **A benign remote environment does not prove local stability. Apollo 13 recorded less than 0.05 g at the spacecraft while the center-engine attachment approached 34 g. Put sensors and models where the loop participates.**
2. **Pogo must be treated like flutter or control instability. A specified vibration level is a response snapshot; stability asks whether the coupled system creates or dissipates energy. Demonstrate margin across operating time and uncertainty.**
3. **Protection and graceful degradation are necessary, not sufficient. The automatic shutdown and engine-out guidance worked. The lasting reliability improvement came from changing the coupled plant with a LOX-line accumulator.**

Reflection

A new upper stage predicts only 0.03 g at the payload at a feedline/structural-mode crossing, but the coupled analysis shows -0.2% damping for one dispersed parameter set. Is the payload result acceptable evidence of safety?

Compact answer. No. The payload response says little about local loop stability, and negative damping means the assumed response is not stationary. Resolve the unstable dispersed case by improving/correlating the model, changing the physical system or operating trajectory, and verifying the loop with local pressure and acceleration measurements. Payload environment compliance can be assessed after stability is established.

References

[1] NASA Manned Spacecraft Center (September 1970). Apollo 13 Mission Report, MSC-02680 / NASA-TM-X-66449. NTRS record. Mission trajectory and spacecraft vibration.

[2] NASA Marshall Space Flight Center, Saturn Flight Evaluation Working Group (June 20, 1970). Saturn V Launch Vehicle Flight Evaluation Report - AS-508, Apollo 13 Mission, MPR-SAT-FE-70-2. Public scan. Primary post-flight event, propulsion and structural data.

[3] Larsen, C. E. (May 5-8, 2008). NASA Experience with Pogo in Human Spaceflight Vehicles, RTO-MP-AVT-152. NTRS record. NASA retrospective and program lessons.

[4] NASA (October 1970). Prevention of Coupled Structure-Propulsion Instability (Pogo), NASA SP-8055. NTRS record. Historical design criteria and recommended practices.

Verified October 3, 2026. Figures 1-3 and tables are original. Figure 2 and all Apollo values are compiled from [1-3]. Figure 3 is an illustrative calculation, not measured flight data. No proprietary program information is used.